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ENTERPRISE SYSTEM-ENABLED ORGANIZATIONAL AGILITY CAPABILITY: A CONSTRUCT AND MEASUREMENT INSTRUMENT

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Abstract

In the turbulent and highly competitive environment, organizational agility (OA that is the ability of organizations to quickly sense and respond to environmental changes, is an important determinant of organization success. Pervasiveness of enterprise system (ES) implementation (e.g. CRM, ERP, SCM, etc.) which aims to integrate information resources in particular and organizational resources in general has raised the questions on its link to organizational agility. Using insights from dynamic capability theory, this paper refers to the capabilities that organizations can leverage from their ES in supporting to organizational agility as the ES-enabled Organizational Agility (ESEOA), and develops a set of variables to measure this construct. The measures were tested for content and construct validity and reliability based on data from 180 medium and large Australian and New Zealand organizations that have implemented and used ES. The research is significant to provide researchers the metrics as part of nomological network of factors to explain the post-implementation impacts of ES on organizational performance. It at the same time provides practitioners feasible tools in analyzing the ES capabilities in their organizations.

Keywords: Enterprise System Competence, Information Systems, Organizational Agility, Dynamic Capability Theory.

1 INTRODUCTION

Organizational agility (OA) which refers to organizations' ability to thrive by sensing and responding to environmental changes has become critically important nowadays when the business environment is getting highly competitive and turbulent. Being regarded as a key business factor and a potential enabler to organization's competitiveness (Mathiassen and Pries-Heje 2006), it has become the topic of interest of both academics and practitioners in recent years. Nine out of ten executives ranked OA as both critical to business success and growing in importance over time in a McKinsey & Company survey (Sull 2009). Research on OA is particularly emerging in information systems field (Izza et al. 2008; Sambamurthy et al. 2003) due to the extensively reliance of contemporary organization on information in general and its information system in particular (Mathiassen and Pries-Heje 2006).

Although there are a number of studies on the relationship between IS and OA, two gaps have been identified from the existing IS literature. Firstly, IS literature has emphasized on IS and its capabilities as the means to obtain organizational agility. However, the literature in regard to how IS contribute to OA is either not conceptually transparent and rather treats the process that connects IS and OA as a "black box" or has not been validated with empirical evidence (Overby et al. 2006; Seo and Paz 2008). Among the limited number of empirical studies on agility such as the research examining the relationship among technical IT capability, managerial IT capability and business process agility in the context of environmental dynamism (Tallon 2008), the research investigating the effect of an organization's IT infrastructure capabilities on enabling IT-dependent organizational agility (Fink and Neumann 2009; Raschke and David 2005), majority of them adopt a factor-oriented approach that overlooks upon the mechanism by which the IT/IS capabilities can be achieved. Other studies although have proposed the framework suggesting how IT-enabled organizational agility capabilities are generated are either conceptual in nature (Sambamurthy et al. 2003) and not supported by empirical evidence (Overby et al. 2006). Recent research on the process of IS development suggests a roadmap that organizations can deploy their IS to be capable in enabling organizational agility (Tan et al. 2010). Although some guidelines in regard to achieving OA from IS perspective can be derived from the research, however, the focus of these studies is on the agile development of IS rather than the role of IS itself. As such, there is a need for a theoretical framework to explicate how IS-enabled organizational agility can be achieved. Secondly, much of the IS and OA literature uses broad and overly abstract theoretical IT-related constructs as the antecedents of OA such as digital options (Sambamurthy et al. 2003), IT usage and IT acceptance (Zain et al. 2005). These studies either lack attention to the attributes of specific types of information systems or use various anecdotal examples that might support the use of the technology or the methodology the studies suggest, thus providing less specific indications for practical action (Orlikowski and Iacono 2001). The above two gaps in IS literature drive this research which is particularly concerned about one set of information systems, Enterprise Systems (ES), and their relationship to OA. ES refers to an integrated IS that uses both technology and management capabilities of that technology to manage information flow in an organization (Davenport 1998). ES are different from other types of IS by the scope and depth of information they manage. ES covers not only internal business units but also interrelationship with business partners. Information managed by ES is at all the levels from operational to strategic level. ES inherit some of the IS characteristics but have unique features such as standardization, integration, process oriented and best practice (Goodhue et al. 2009; Lengnick-Hall et al. 2004) which results in distinctive contribution towards the organization performance in comparison with other types of IS such as decision support system, or transaction processing system. ES such as Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and Supply Chain Management (SCM) harness the power of contemporary IT and are pervasively used in most large organizations (Davenport 1998).

Although most of IS researchers studying on organizational agility have earlier referred to IS with implication of enterprise system, or taking ES as an example of IS, the lack of theories specifically about IS artifacts (its context, effects, capabilities) in those studies limits the research contribution to the understanding of IS roles in organization agility (Orlikowski and Iacono 2001). Therefore,

research on the role of ES on the organizational agility is expected to fill the above gaps from the literature.

Despite the pervasiveness of ES and the importance of OA, the relationship between ES and OA remains under researched. We conducted a thirteen-year (1998-2010) review of papers published in the top 10 IS journals. The finding reveals that there are only five academically published papers on ES and OA (Davis 2005; Gattiker et al. 2005; Ignatiadis and Nandhakumar 2007; MacKinnon et al. 2008; Seethamraju and Seethamraju 2009). Given the trends that most large organizations are moving towards ES and are investing heavily on ES architecture and the increasing demand for an agile organization, lack of research on ES and OA is a critical oversight. Further analysis and syntheses of the previous ES, IS and OA literature identified additional gaps. Firstly, most ES research focuses on the implementation issues of ES and not how it either inhibits or limits OA (Moon 2007). The exceptions are the works of (Davis 2005; Gattiker et al. 2005; Ignatiadis and Nandhakumar 2007; MacKinnon et al. 2008; Seethamraju and Seethamraju 2009). However, in these studies organizational agility is not a dependent variable. Instead, they research on the influence of ES on business process agility (Seethamraju and Seethamraju 2009), system agility (Davis 2005), and organizational resilience (Ignatiadis and Nandhakumar 2007). Furthermore, the majority of the research is conceptual research, and among the empirical research, the majority employs qualitative research methodology (Seethamraju and Seethamraju 2009). Some published papers are working papers and both their conceptual framework and findings are yet to be confirmed (Davis 2005). Secondly, there is contradiction in the role of ES and OA from the limited available research. Some authors claim that ES has a negative impact on agility because of tight integration to make any process change (Newell et al. 2007), lack of functional fit with business requirement (Ni et al. 2002). Conversely, (Gattiker et al. 2005) state that ES contain some mechanisms such as built-in flexibility, process integration, data integration, and availability of “add-on” software applications by which to support agility. Thirdly, the recent introduction of new ES technologies such as SOA and web services have altered the ES infrastructures which lead to the changes in ES capabilities (High et al. 2008; Macvittie 2006). The aforementioned gaps provide justifications for conducting a study that aims to investigate the impact of ES on organizational agility.

Drawing from dynamic capability theory and process based theory, our previous research (Trinh-Phuong et al. 2010) proposed a conceptual framework to explain how ES enables organizational agility (Figure 1). In that framework, ES competences do not directly impact on agility level. Instead, through the process of assimilating ES in organizations’ activities, ES competence is leveraged to developed dynamic capabilities which we refer as enterprise system-enabled organizational agility capability (ESEOA). ESEOA in turns is embedded in the two fundamental business environment sensing and responding processes of organizational agility. Hence, ESEOA is the missing link between ES competence and OA.

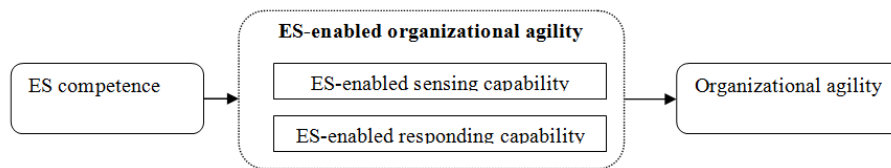


Figure 1. Framework explaining how ES competence enables organizational agility

The current research, which is successive to that research, reports the results of a study that further examines the theoretical meaning, structure and dimensionality and reliability and validity of construct enterprise system-enabled organizational agility (ESEOA). The paper is expected to provide potential measurements indicators of ESEOA, thus, it helps to advance a further step in validating the framework that we propose earlier in explaining how ES impacts on OA. The rest of the paper is structured in the following sessions: Section 2 discusses on the relevant literature; Section 3 presents the methodology pursued to develop the measurement instrument and how we operationalize the methodology in developing the measurement instrument. Section 4 provides a brief discussion of the results of the exploratory and confirmatory factor analysis. Finally section 5 discusses the theoretical and practical implication of the findings together with limitation and future research possibilities.

2 CONCEPTUAL BACKGROUND

The conceptual foundation of enterprise system enabled organizational agility (ESEOA) is derived from three research domains: organizational agility, dynamic capability theory, enterprise system and information system competence. While the dynamic capability theory helps us understand the transformation of ES resources into ES capabilities in the unstable business environment, the OA literature identifies the processes that ES can be used in enhancing the OA level and the literature on ES competence help to identify what ES competence are relevant to be leveraged in supporting OA.

2.1 Organizational Agility

The concept of OA is derived from performance characteristics of an agile organization and is rooted in two related concepts- “organizational adaptability” and “organizational flexibility” (Sherehiy et al. 2007). Organizational adaptability focuses on how an organization’s form, structure, and degree of formalization influence its ability to quickly adapt to its business environment (Sherehiy et al. 2007). Organizational flexibility represents an organization’s capacity to adjust its internal structures and processes in a predetermined response to changes in the environment. Adaptability underlies the fit of organizational operations to their environment while flexibility emphasizes the readiness of organizational resources and the ease of resource mobilization. The “agility” concept encompasses both flexibility and adaptability (Christopher and Towill 2001). An agile organization is not only “flexible” to cater for predictable changes but also is able to respond and adapt to unpredictable changes quickly and efficiently (Oosterhout et al. 2006). Therefore, OA can be viewed as the state of organizational performance in terms of flexibility and adaptability and is attainable through organization’s activities. In particular, from the process-based perspective, OA is a set of processes that allow an organization to sense changes and respond efficiently and effectively in timely and cost-effective manner in the internal and external environments (Seo and Paz 2008). Sensing refers to an organization’s ability to *detect, capture and interpret* organizational opportunities (Oosterhout et al. 2006; Seo and Paz 2008). Responding represents an organizational ability to mobilize and transform resources to react to the opportunities that it senses (Gattiker et al. 2005; Oosterhout et al. 2006). These two capabilities must be aligned to optimally obtain OA (Overby et al. 2006). Organizations which have high level of sensing but low level of responding will not be able to take the opportunities for better performance while organizations which have higher level of responding capabilities but low level of sensing capabilities would waste their resources. Any impact upon the environment sensing and responding capabilities of an organization would consequently affect the OA level.

2.2 Perspective of Information System and Enterprise System impacts on Organizational agility

IS literature agrees upon the general benefits of enterprise systems including (1) integrating data and applications, implementing generic business model based on the “best practices”, (2) providing standardized solutions of the business problems, (3) building on modularized structure, and allowing customizations for the specific business process (David et al. 2003; Lorincz 2007). These general benefits were further analyzed from organizational performance perspectives and acknowledged in 5 areas: operational, managerial, strategic, IT infrastructure and organizational benefits (Shang and Seddon 2002). Although OA are highly correlated to the business performance, the literature is divided in three distinctive views – *facilitating, inhibiting and neutral views* - in regard to the impact of IS in general and ES in particular on OA. Some authors claim that ES has a negative impact on agility because of tight integration to make any process change and lack of functional fit with business requirement (Ni et al. 2002). Customization in ES, which results from the lack of fit between organizationally owned business processes with standard processes provided by ES vendors, brings in complexity which may reduce OA (Davis 2005). Conversely, other research suggests a positive impact of ES on OA. An increase in OA is one of the benefits brought about by ES implementations (Davenport 1998). Firstly, due to the advancement of modern IT, the impacts of ES on OA can be identified in three areas including speeding up activities; providing intelligent and autonomous

decision-making processes; and enabling distributed operations with collaboration which enable agility (Huang and Nof 1999). Secondly, ES contain mechanisms such as built-in flexibility, process integration, data integration, and availability of “add-on” applications to support agility (Gattiker et al. 2005; Goodhue et al. 2009). Vertical integration which is resulted by ES implementation enhances the visibility, accessibility and control and decision support capability, thus enables top management to comprehend better the critical need for changes in supporting agility (Seethamraju 2009). (Huang and Nof 1999) also argue that if the definition of agility is concerned with an ES which consists of multiple, inflexible but cooperative subsystems, the whole system may still provide high agility. Through the availability of built-in and add-on functionalities, which are developed to serve special requirements, ES provide more options for responses to meet agility challenges that do not require changing the backbone of tightly integrated program code (Goodhue et al. 2009). Hence, built-in and add-ons solutions can be viewed as one sources of agility provided by ES. Moreover, ES vendors are reflecting on the market’s demand to innovate their software products by providing new version of ES systems (for example SAP’s ERP system has evolved from R3, through ECC6 to business suite 7.0). This is an incremental and continuous development without radical modification on the basic concept of ES. At present, the ES eco-system is undergoing through a paradigm shift. On the business side, agility and flexibility are becoming essential features of modern day organizations. On the technology (IT) side, two architectures have emerged emerging: the web based SOA (service oriented architecture) and BPMS (business process management suites) (Lorincz 2007). The introduction of new ES technologies such as SOA, business process management (BPM) and web services have altered ES infrastructures, which lead to changes in ES capabilities (Moitra and Ganesh 2005). Loose coupling which is the fundamental technique of SOA has been embedded into the development of the most recent ES, hence, making ES better agility-enabling applications (Chen et al. 2007). Lastly, some researchers have introduced a neutral conclusion on the roles of ES on organizational agility. The introduction of an ES creates power differentials, which serves to increase control in the organization. This results in increased rigidity, and a possible decrease in organizational flexibility and resilience. On the other hand, ES can also cause drift, resulting from the unexpected consequences of this power differential, as well as from the role of perceptions of people in solving a problem within the ES. This reduction in control may serve in some circumstances as an enabler of organizational resilience (Ignatiadis and Nandhakumar 2007).

We argue that the contradictory conclusions on the relationship between ES and OA in the existing literature are due to the lack of mechanism that explains the impact of ES on OA. The previous studies mostly look into IS in general and ES in particular as resource providers. However, as (Shang and Seddon 2002) suggest that ES benefits are indirectly linked with business benefits through the use of information, rather than ES itself, ES mismanagement is the main reason that negatively influences OA (Overby et al. 2006).

2.3 Dynamic capability theory

The DCT regards firms’ ability to constantly adapt, renew and reconfigure their capabilities and competences as the major source of performance (Teece 2007). Dynamic capabilities are “the organizational and strategic routines by which firms achieve new resource configurations as markets emerge, collide, split, evolve, and die” (Eisenhardt and Martin 2000). While RBV emphasizes the appropriate selection of resources, DCT emphasizes the evolvement of resources (Teece 2007), which is signified by two processes: resource-picking and capability-building in the organization-learning loop. Hence, from the DCT perspective, organizational resources need to be adaptive, renewable and reconfigurable to provide sustainable competitive advantage (Teece 2007). Dynamic capabilities are commonly associated with dynamic environment where an organization needs to keep changing its resources to suit the organization strategy at a particular circumstance (O'Connor 2008). Under different conditions of business environment, different types of dynamic capabilities are needed. The DCT provides a relevant theoretical lens to conceptualize the link between ES and OA. ES can represent valuable and arguably rare resources. They provide the essential platform; can be key sources of organizational capabilities and can potentially, albeit indirectly, contribute to OA (Sambamurthy et al. 2003). However, ES resources are not static or ultimately superior in enabling

OA. Instead, ES resources need to be carefully selected, configured and combined with other non-ES organizational resources to generate ES-enabled organizational agility capability for OA. Since sensing and responding capabilities are two fundamental capabilities of an agile organization, thus, ES-enabled organizational agility capability is a dynamic capability and reflected by the extent of ES-enabled sensing capability and ES-enabled responding capability.

Table 1 specifies the construct domains and relevant concepts extracted from the literature that the ESEOA can be conceptualized from.

Concept	Definition	Relevance to the current research
Organizational agility	The performance of an organization to excel in utilizing its resources in order to quickly sense changes from its business environment and respond to those changes appropriately.	Identify ES capabilities enabling sensing and responding changes
Sensing capability	Organization's ability to detect, capture and interpret organizational opportunities	Identify ES capabilities that supports organizations in capturing, interpreting, and providing business intelligence
Responding capability	Organizational ability to mobilize and transform resources to react to the opportunities that it senses	Identify ES capabilities that supports organization in mobilizing resources, and taking actions
Dynamic Capability Theory	Organizational resources need to be adaptive, renewable and reconfigurable to provide sustainable competitive advantage	Help identifying ES capabilities that are adaptive, renewable, and reconfigurable

Table 1. Summary Definition of Key Concepts

3 SCALE DEVELOPMENT AND VALIDITY

There is an alarming lack of effort in validating instruments (Straub et al. 2004) address a relative paucity of replication in MIS. To avoid the unstructured construct define, the research applies the three-stage construct development proposed by (Lewis et al. 2005).

3.1 Stage 1: Domain definition of the “Enterprise System-enabled organizational agility capability ESEOA”

The first state is to establish the domain of the idea whereby the purpose and/or importance of the construct, the conceptual definition, and the list of dimensions which represent the elements of the constructs are specified. The concept ES-enabled organizational agility (ESEOA) is drawn upon the concepts of ES competence transformation into capabilities from DCT perspective. This study argues that ESEOA is generated when ES competence is leveraged in the process that organization senses and responds to business changes. Thus, ESEOA are derived from sensing and responding capability. Since changes happen in business environment unpredictably, the leveraging of ES competence in enabling organizational sensing and responding capabilities, which results in the generation of ESEOA, is a continuous process. Based on the review of enterprise system literature and dynamic capability theory, the ESEOA is defined as: “*organization's capability to deploy its ES resources and embed them in the organization's activities to quickly and efficiently sense and respond to changes*”. Hence, ESEOA is a higher order concept composed of two constructs ES-enabled sensing capability (ESESC) and ES-enabled responding capability (ESERC) (Figure 2).

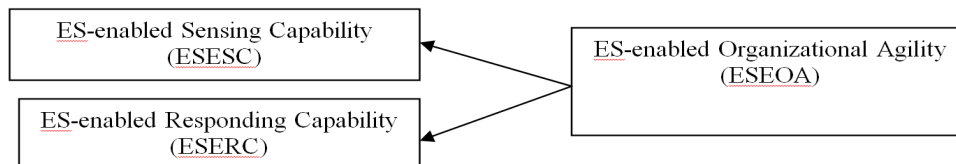


Figure 2. ES-enabled Organizational Agility

3.1.1 ES-enabled Sensing capability

Sensing capability focuses on a firms' ability to gather and make sense of externally generated knowledge to manage disruptive changes (Overby et al. 2006). Sensing capability not only indicates the ability of an organization to sense current changes but also to develop market foresight to anticipate changes in the future. Organizations that are able to anticipate changes in their business environment can quickly devise their responding actions in advance of their competitors. There are various ways of building sensing capability. (Neill et al. 2007) argued that organizations that perform a better capability to *communicate* relevant information among members of the decision making team, *interpret* their environment in a multidimensional way and analyze the information simultaneously by incorporating multiple perspectives will have higher sensing capability and eventually become more agile. Furthermore, *anticipatory capability*, which refers to the ability to predict the way that market is moving can be an essential dimension of sensing capabilities (Day 1994). Overall, the development of sensing capability requires organizations to scan the business environment and capture business insights beyond the usual sources. Such capability can be developed by organizational technologies, processes, values and norms that together generate knowledge about the future condition (Sambamurthy et al. 2003). Based on the above logic, we postulate that ES, as valuable resources, can be deployed as a source of capability building mechanisms to either directly or indirectly enable sensing capability. We name this construct as *ES-enabled sensing capability* and define it as “*ES-enabled sensing capability refers to the ability of an organization to quickly and efficiently utilize its ES to digitize the process of sensing and develop a strategic market foresight about its business environment*”. The sensing process includes capturing, interpreting and analyzing change signals from business environment. ES functionalities can digitize the sensing process.

3.1.2 ES-Enabled Responding Capability

While sensing capability generates knowledge of the business environment, responsive capability effectively transforms that knowledge into action (Gattiker et al. 2005). Response capability is thus reflected by the change-enabling capabilities that are embedded in organizational processes (Li et al. 2008). (Christopher et al. 2004) suggest responsiveness to be characterized by a short time-to-market, the ability to scale up (or down) quickly and the rapid incorporation of consumer preferences into the design process. Response acts are the result of a range of operating and strategy capabilities that organizations develop. (Overby et al. 2006) suggest four fundamental strategic capabilities: (1) *production development capabilities* to facilitate a firm's ability to embark on new ventures; (2) *systems development capabilities* to quickly and efficiently implement change to existing systems such as reusable service, SOA; (3) *supply-chain and production capabilities* to adjust existing ventures by shifting production to match a pending change in demand such as high supply chain visibility; and (4) *flexible resource utilization* to shift resources to areas of need to embark on new ventures or adjust existing ventures. Based on the above understanding of responding capability, we postulate that ES as valuable resources can be deployed as a source of responding capability building mechanisms. We name this construct as *ES-enabled responding capability* and define it as following with table five summarizing the concept matrix: “*ES-enabled responding capability refers to an organization's capability to deploy its ES resources and embed them in its production development, systems development, supply chain and production and flexible resource utilization strategies and processes to quickly and efficiently respond to changes*”.

Based on the above definitions, content analysis to draw inferences from text to address the domain specified above from the literature and the discussion with two domain experts, 25 items including 9 items (ESS1-9) to measure ES-enabled Sensing capability (ESESC) and 16 items (ESR1-16) to measure ES-enabled responding capability (ESERC) were generated (Table 2). These items are measured on 5-point Likert scale.

Construct	ID	Items
ESESC	ESS1	Capture business intelligence from various sources (customer, competitor, supplier)
	ESS2	Generate knowledge about the market (<i>market trend, competitors' actions, regulation changes, cultural shifts, technology developments, etc</i>)
	ESS3	Interpret business intelligence for different management levels
	ESS4	Analyse business intelligence in different formats (text, audio, video)
	ESS5	Prioritise the most important changes in the business environment
	ESS6	Provide business intelligence to decision makers of different functional units across the organization
	ESS7	Develop real time visibility of demand in your supply chain
	ESS8	Develop alertness about the business environment
	ESS9	Develop rich industry foresight
	ESS10	Examine trends in the data for the industry foresight
ESERC	ESR1	Quickly bring new products/services to market
	ESR2	Quickly add more feature (s) to existing products/services
	ESR3	Introduce new product/service faster than other competitors
	ESR4	Easily adjust the production volume of products/services
	ESR5	Create a high degree of process interconnectivity with trading partners
	ESR6	Simultaneously work on the same data with trading partners
	ESR7	Collaboratively plan with trading partners
	ESR8	Allow trading partners to work on your real data
	ESR9	Increase the accuracy of the data used by trading partners
	ESR10	Simultaneously develop information systems with several supply chain partners
	ESR11	Simultaneously design business processes with several supply chain partners
	ESR12	Increase the accuracy of information used by top management in making strategic decisions
	ESR13	Create a high degree of intra-organizational business process interconnectivity
	ESR14	Adapt to radical market changes
	ESR15	Empower end-users for taking actions in business operation
	ESR16	Resilient to radical changes
	ESR17	Generate new business strategies

Table 2. Initial items for ES-enabled Organizational Agility Capability

3.2 Stage 2: Instrument purification

To purify the instrument and establish content validity, these measurement indicators are tested through pretesting and pilot testing. The pretesting was conducted with a panel of experts comprised of 15 professors who have published researched on agility and its closely related concept such as flexibility, adaptability in high ranking IS journals. The panel of expert were asked to complete the questionnaires and rate on the degree of relevance of each items on measuring the underlying constructs on the scale from 1 to 5 with 1 is “not relevant” and 5 is “highly relevant”, as well as to provide additional comments on every aspect of the instrument such as format, content, comprehensiveness and ease of completion. To assess the consistency among different experts, the inter-rater correlation was calculated. At $p=0.01$, all of the inter-rater were significant, thus supporting the reliability of the experts judgment. Items with a mean relevance score of less than 3.5 were dropped. Items ESS4, ESS8, ESS9 of ESESC and items ESR2, ESR3, ESR6, ESR8, ESR11, ESR14, ESR16 of ESERC were deleted due to that criterion. Item ESS5 from ESESC and items ESR1, ESR7, ESR12 were changed in wording. Two items ESS10 "Examine trends in the data for the industry foresight" and ESR17 “Generate new business strategies” were added after being suggested from the panel of experts and the literature review.

After the modification of the questionnaire was done based on the review of the feedback from the panel of experts, questionnaire was further tested by a pilot testing. Three respondents were chosen from the same population from which data would be collected. They were asked to complete the survey questionnaire and give suggestion for improvement on the survey. The pilot study aimed to ensure the comprehensiveness of the items, the appropriateness of scale used in measuring items and

the contextual appropriateness of language. Changes on the wordings of the items were made after the pilot test. The steps taken in stage one and two ensure the content validity of the instrument. The survey questionnaire after stage two is the final draft of the instrument which we would bring to stage three. ESESC were measured by 7 items and ESERC were measured by 10 items.

3.3 Stage 3: Data collection

In the third stage, we conducted a survey on medium to large Australian and New Zealand organizations that have implemented and used enterprise systems for at least 1 year. We focused on three main enterprise systems: ERP, CRM and SCM. The invitation to take part in to the survey were sent to 1400 CIOs (or equivalently high ranking IT manager) of organizations that were randomly selected and provided by specialist data provider and gather from a variety of industry sectors to make the result generalisable.

There were a total of 275 emails bounced back and 139 respondents requested to be excluded in the research. The reason for the bounce back email was either wrong address, or the respondents have moved or changed their post. This made a final sample size of 986 respondents. Out of 224 responses received which made a response rate of 22.7%, 180 responses were included in the final dataset and retained as usable for this research after excluding 44 incomplete responses in a meticulous data cleaning process. This response rate is in line with the response rate for studies on senior IT managers in large organizations in literature.

4 DATA ANALYSIS

A preliminary data analysis was undertaken to examine the psychometric properties of the scales via measures of central tendency, correlation reliability estimates, exploratory factor analysis and confirmatory factor analysis (CFA) that we will discuss in the following sessions.

4.1 Nonresponse bias and Common method bias

The wave analysis (Armstrong and Overton 1977) is applied to test the nonresponse bias by comparing the responses from pre-deadline survey with the late responders on actual survey variables. This method assumes that people who respond in later waves to have responded because of the increased stimulus and are expected to be similar to non-respondents. Therefore, if the late respondents are different from respondents, it most likely suggests that some level of bias exist. We tested for statistically significant difference in the responses of the 72 late respondents to the earlier on all the variables using a one-sample t test. The result indicates the significant values were all larger than 0.05. Thus, even if non-response bias cannot be excluded, it is not statistically significant.

Common method bias may occur from data that is collected only via one method or only at one point of time (Straub et al. 2004). Among several methods to test common method bias, this research applies Harman's single factor test which is most widely used (Podsakoff et al. 2003). An exploratory factor analysis was conducted upon the full data set. If a substantial amount of common method variance is present either (a) one factor accounts for the majority of the covariance between the measures or (b) a single factor will emerge from the factor analysis (Podsakoff et al. 2003). The unrotated factor solution results in two factors with no general factor accounting for the majority of the variance.

4.2 Unidirectional Validity with Exploratory Factor Analysis (EFA)

Since, the ESEOA construct was hypothesized as the higher-order construct composed of two constructs ESESC and ESERC, hence, EFA is used to derive the initial set of factors for the construct (Lewis et al. 2005). An important concern in EFA is sample adequacy. (Hair et al. 2010) suggested a case-to-variable ratio of 5:1 to guarantee a reliable EFA procedure. The measurement contained 17

variables with a sample size of 180. Therefore the sample adequacy criterion was met. In order to verify that the data are meant to factor analysis, two tests should be executed: the Kaiser-Meyer-Olkin (KMO) test and the Bartlett sphericity test. The value of KMO is 0.917 which is larger than the acceptable value of 0.5. The significant value ($p < 0.001$) indicates that the correlation matrix generated from the data is different from the identity matrix. Therefore, the factor analysis is appropriate in this research.

In EFA, Principal Component Analysis (PCA), Varimax rotation method, and minimum Eigen value of 1 were chosen as the condition for factor to be extracted. The threshold of item loading for EFA used in IS construct development studies are 0.5 (Lewis et al. 2005). Items that have no loadings exceeding the threshold should be dropped from further analysis. As the result, two factors were identified (Table 3). This result is consistent with the hypothesis that ESEOA is composed of two constructs ESESC and ESERC. EFA was conducted with several rotation techniques to reach the best factor solution based on the judgment on the simplicity, interpretability and the percent of variance explained by the set of designated factors. The results yielded the same numbers of factors as well as the component variables. With reference to the item with largest loading and the overall nature of the items that loaded on the factors, these two factors were labeled as ES-enabled Sensing Capability (ESESC) and ES-enabled Responding Capability (ESERC) respectively. Item ESR13 “Create a high degree of intra-organizational business process interconnectivity” was dropped because of cross-loading. Three items (ESR12, ESR15, ESR17) that are theorized to measure responding capability loaded with sensing capability. These items refer to increasing access to information and empowering decision makers to take strategic actions. Respondents might have considered this as part of their sensing process. The two factor model account for 58.4% of the total variance to be explained.

	ESESC	ESERC
ESS3	.865	
ESS6	.823	
ESS5	.775	
ESS2	.715	
ESS10	.709	
ESR12	.696	
ESS1	.695	
ESR17	.611	
ESS7	.586	
ESR15	.562	
ESR10		.824
ESR5		.819
ESR7		.762
ESR9		.635
ESR4		.631
ESR1		.505

Table 3. Factor structure after Exploratory Factor Analysis

4.3 Construct validity with Confirmatory Factor Analysis (CFA)

The construct validity is conducted investigate if the instrument items selected for a given construct are a reasonable operationalization of the construct (Straub et al. 2004). The validity of the measurement model of ESEOA was tested using CFA using AMOS. The univariate normality test was conducted on each of the variables to confirm the multivariate normality requirement for CFA. The CFA was conducted in the order of (a) measurement properties of each factor separately, then (b) full factor measurement model (first-order and second-order) of the ESEOA construct to assess overall fit and (c) construct validity.

4.3.1 Unidimensionality

Firstly, each of the factors identified from the EFA were independently evaluated in a measurement model CFA to test if it fits the data (Figure 3a-3b). In the process of fitting the first order models, four items were dropped from the ESESC because of poor convergent validity. These are all the three items (ESR12, ESR15, ESR17) that were supposed to measure the ESERC but that loaded to ESESC in the EFA, and ESS7. In addition, one item (ESR7) was dropped from the the first order model of the ESERC because of poor convergent validity. The Goodness-of-Fit (GOF) shows that the two measurement models satisfy construct validity and fit the data reasonably.

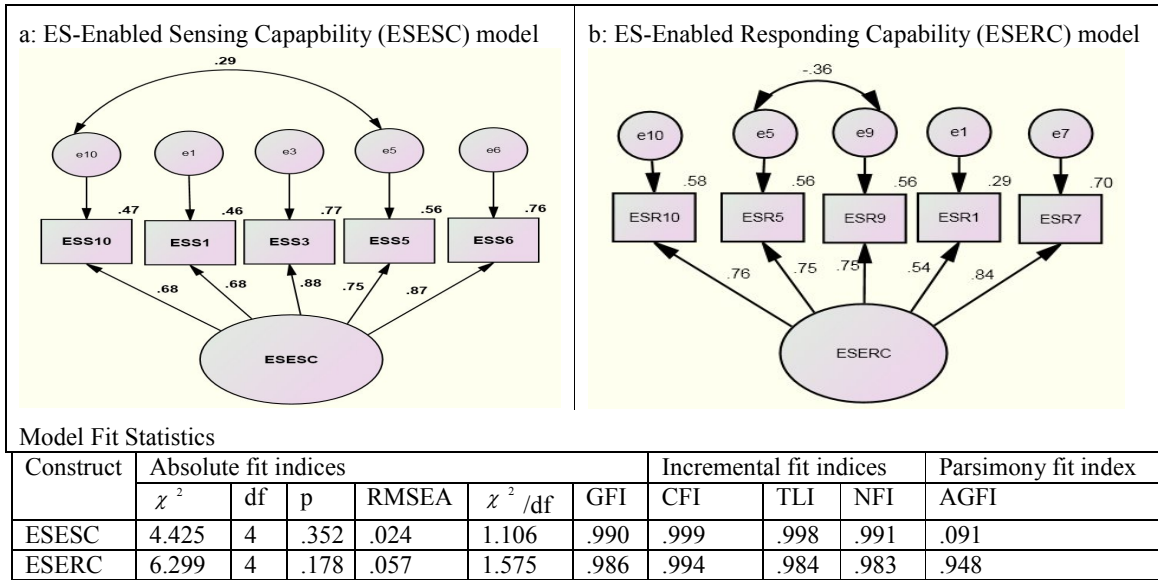


Figure 3a-3b. Factor models of the ESEOA construct

4.3.2 Full-measurement model of the ESEOA construct

Once the unidimensionality of each of the factors of ESEOA construct was set up, the entire measurement model of the ESEOA construct was evaluated if it fits the data. The full first order then the second order measurement model was tested along with the validity indices of the constructs-convergent validity and discriminant validity. The full first order model for ESEOA is a two-factor-eight-item model. Once the first order model is derived we proceeded to establishing if ES-enabled sensing and responding capabilities share the same domain and fall within the nomological net of the ESEOA through a second order model of ESEOA. The result (See figure 4a-4b) shows the CFA output of the full first-order and second-order measurement model respectively. The GOF indices indicate that the propose model fit the data reasonably. Reviewing the modification indices (MIs) and regression weight suggest no problem with the models.

4.3.3 Construct validity

Once the GOF of the proposed measurement model for ESEOA is established, the construct validity is assessed. Construct validity is the extent to which a set of measured items actually reflects the theoretical construct those items are assigned to measure (Hair et al. 2010). For this study, convergent and discriminant validity and reliability for the final measurement model are estimated based on the factor loadings, SMC, AVE and Construct Reliability. Convergent validity assesses if the indicators of a specific construct should converge or share a high proportion of variance in common. Table 4 summarizes the results of the convergent validity evaluation. The results were all higher than the threshold value which indicates that there is convergent validity of the ESEOA construct.

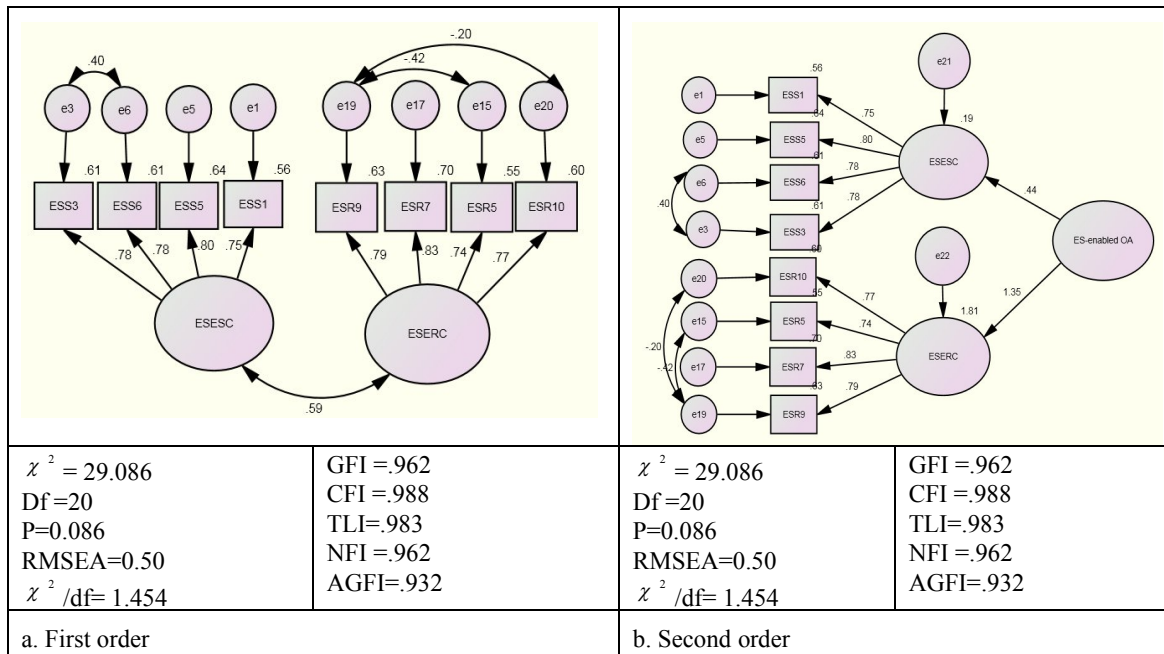


Figure 4: Final full first-order and second-order measurement models

Variable	Item	SFL	SMC	AVE	Construct Reliability	Cronbach's Alpha
ESESC	ESS3	.781	.609	0.604	0.8	0.872
	ESS6	.778	.606			
	ESS5	.802	.643			
	ESS1	.747	.558			
ESERC	ESR9	.791	.625	0.616	0.6	0.841
	ESR7	.835	.697			
	ESR5	.738	.545			
	ESR10	.772	.596			

CR(.6 or higher), AVE (.5 or higher), SFL (.5 or higher) SMC (threshold .3 or higher)

Table 4: Validity and Reliability Analysis of Measurement Model

The discriminant validity measures the extent to which a construct is truly distinct from other construct. It is assessed through the comparison of the AVE to inter-construct squared correlation estimates associated with that factor (Hair et al. 2010). The result (Table 5) shows the AVE values are greater than their respective square correlation which indicates a good evidence of discriminant validity.

	ESESC	ESERC
ESESC	.604	.498**
ESERC	.498**	.616

**. Correlation is significant at the 0.01 level. Diagonal elements (value in bold font) = AVE, off-diagonal elements = squared inter-construct correlations.

Table 5: Discriminant validity of the two factors (ESESC, ESERC) of the ESEOA construct

5 DISCUSSIONS

Although several studies have tried to link ES with OA, they mostly stop at identifying potential enterprise system competence that can impact on the agility level. None of these papers give a rigorous explanation of how the impact caused by ES-associated resources is performed. This paper, by developing ESEOA concept from DCT perspective addresses the gap. ESEOA is defined as a

higher order construct composed of ESESC and ESERC subconstructs. The paper also developed an 8-item measurement instrument to assess ESEOA with each 4 items to measures ESESC and ESERC respectively. ESESC is measured by ESS1, ESS3, ESS5, ESS6 suggesting that organizations which excel in using ES in capturing business intelligence from various sources, interpreting it for different management levels, and providing it to decision makers of different functional units across the organization while prioritizing the most important changes in the business environment will be able to quickly sense the changes from business environment. Likewise, ESERC is measured by ESR5, ESR7, ESR9, ESR10 indicating that organizations would be able to quickly respond to change if they can use their ES to create a high degree of process interconnectivity with trading partners, collaboratively plan with them while at the same time increase the accuracy of the data used by trading partners and simultaneously develop information systems with them. The measurement of ESESC and ESERC suggests that while ES enable sensing capability of organization in all extents ES only support responding capability through enhancing the relationship with its partners and not via its contributions to the organization's internal activities. Thus, ES that focuses in internal organizational operation like ERP would only contribute to organizational agility in terms of supporting sensing capability while it supports organization agility limitedly in terms of responding capability. In order to achieve organizational agility from ES, organizations are required to expand their ES beyond the organization scope.

The measurement model measurement scale demonstrates acceptable psychometric properties and fit the data. The ESEOA construct was developed by following strictly and systematically instrument design guideline suggested by (Lewis et al. 2005). The content validity was ensured by following the a rigorous process of item development based on theoretically grounded definition of ESEOA, comprehensive literature review, consultation with panel of experts survey and pilot testing. The construct validity was ensured by following a systematic test of factorial validity, convergent validity and discriminant validity. The measurement model demonstrates adequate psychometric properties and fit the data which indicates that the instrument is ready for use in empirical research to represent the dynamic capabilities that an organization can make out of its ES in sensing and responding to changes.

Some initial items were dropped through out the construct development process (only 8 items in the final measurement model instead of 26 items initially). While this result is understandable since there is paucity of rigorous constructs in the literature that relates to ESEOA, future research should continue to further improve the measurement of the ESEOA construct.

6 CONCLUSIONS, LIMITATIONS AND IMPLICATIONS

The objective of this paper is to develop and validate instrument of measuring ES-enabled Organizational Agility capability (ESEOA). The construct is developed by following the 3-stage instrument development methodology by (Lewis et al. 2005). The domain of construct was defined to anchor the initial construct dimensions and items that operationalize the construct from the relevant literature. The instrument was later purified through consultation with panel of experts and pilot testing. The revived instrument was used to collect data from 180 Australian medium and large organizations in various industries which have implemented and used ES. The EFA and CFA on the data were conducted to develop ESEOA constructs that are composed of two subconstructs ESESC and ESERC. The construct validity of the measurement model was tested using CFA factors. The results showed that the measurement model is valid and reliable both empirically and theoretically.

The study makes a number of contributions to research and practice. Firstly, the current missing link on the mechanism that IS in general and ES in particular impact on OA can be explained by the development of ES-enabled OA construct. While the existing research on the impact of IS on OA is mainly conceptual (Overby et al. 2006; Seo and Paz 2008), this research is one of the few earliest studies that attempt to test the relationship between IS and OA empirically. Furthermore, the measurement instrument of ESEOA construct can be applied in various different and similar contexts to explore the capability of ES in organizational performance, particularly in enabling the ability of organizations to thrive in dynamic business environment. Secondly, this construct also provides an

effective dimension to empirical studies in measuring the post-implementation benefits of ES which are still less researched (Moon 2007). Thirdly, the study adds empirical support to the application of DCT in measuring ES capabilities while majority of the current research only looks at ES from RBV as a source of resources. Fourthly, through the development of two subconstructs ES-enabled sensing capability (ESESC) and ES-enabled responding capability (ESERC), the study not only contributes to the current body of knowledge on how the adoption and assimilation of ES in business processes (i.e. sensing and responding) could generate dynamic capabilities that support OA, but also provides practitioners specifically what ES-enabled organizational agility capabilities are so that organizations could manage their enterprise systems better. Lastly, the study provides practitioners a different view on their ES resources and a benchmark against which they can assess their use of ES in leveraging the OA level. Furthermore, the research findings suggest that in order to develop organizational agility out of their ES resources, organizations should implement ES that support their relationship with the external business partners such as CRM, SCM, etc.

Despite the above contributions, the study has some limitations that need to be taken into consideration and open for future research. The measurement instrument developed in this study has only tailored to organizations in Australia and New Zealand. Thus, testing the instrument in different geographical areas may suggest different result. To the best of the researcher's knowledge, this is first research attempt to quantitatively measure ESEOA, different studies in this area would provide more rigorous theoretical background and solidify the instrument.

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